The Fundamental Structure of the Observational Incompleteness Framework

From Finite Bijection to the Standard Model

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Abstract

The Observational Incompleteness (OI) framework derives quantum mechanics and general relativity from a deterministic bijection on a finite lattice with a visible/hidden partition [1]. This paper addresses two questions the derivations leave open: what is the lattice, and which quantum field theory does the embedded observer see?

For the first question, we show that the framework's results depend on six structural properties — deterministic bijectivity, finite boundary entropy, bounded coupling degree, statistical isotropy, non-trivial partition coupling, and slow-bath capacity — and on nothing else. The regular lattice, the alphabet size q, and the wave equation are all either derived from these properties or irrelevant to the predictions. The lattice is the coupling graph of the bijection: the adjacency structure defined by which degrees of freedom affect which others in one dynamical step. The product decomposition of the state space into "sites" — and hence space itself — is determined by the bijection as the unique factorization minimizing coupling degree. The alphabet size q is a gauge freedom. The spatial dimensionality d = 3 is selected by four independent self-consistency filters. The fundamental ontological commitment is minimal: a finite set, a bijection, and a partition. Everything else — space, time, dimensionality, quantum mechanics, general relativity — is emergent.

For the second question, we show that the wave equation on a d = 3 lattice uniquely determines the Standard Model's structure. Bottom-up: the wave equation factors into staggered Dirac fermions; center independence enforces chiral symmetry, mandating the Higgs; staggered tastes give three fermion generations. Gauge emergence: coupling-degree minimization gives K = 2d = 6 internal components; the cubic rotation group fixes multiplicities (3, 2, 1); background independence promotes the commutant to local gauge invariance. Top-down: anomaly cancellation uniquely determines the hypercharges; the trace-out makes SU(2) chiral while SU(3) remains vector-like; T-invariance forces θ = 0 at all energy scales.

The reconstruction theorem establishes the converse: observed QM with Bell violations, finite boundary entropy, and spatial isotropy uniquely determine the equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$ at the lattice level — a finite set with a bijection of bounded coupling degree and statistical isotropy. Whether (S, φ) *is* reality or *describes* reality is provably undecidable by any measurement — the framework identifies this as gauge.

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1. Introduction

The companion paper [1] proves that an observer embedded in a deterministic system with a coupled, slow, high-capacity hidden sector necessarily describes the visible sector using quantum mechanics — and that these conditions are necessary and sufficient. It also shows that the cosmological horizon realizes these conditions, determines $\hbar = c^3\varepsilon^2/(4G)$, recovers the Bekenstein-Hawking entropy, and dissolves the cosmological constant discrepancy.

Both papers use explicit constructions: a cubic lattice in d dimensions, an alphabet $\mathbb{Z}/q\mathbb{Z}$ at each site, nearest-neighbor coupling, a specific update rule. These constructions work — every link in the derivation chain is analytically proven. But they raise two questions. First, what is the ontological status of the lattice? Three readings are possible: literal (the lattice is physical reality), effective (the lattice approximates unknown structure), and structural (the physics lives in the equivalence class of systems sharing the lattice's structural properties, not in any particular representative). This paper argues for the structural reading and develops its consequences.

Second, does the lattice structure determine which quantum field theory the observer sees? The QM derivation tells the embedded observer that they must use quantum probability; it does not tell them which particles exist or which forces act. We show that the wave equation on a $d = 3$ lattice, combined with mathematical consistency, uniquely determines the Standard Model's gauge group, matter content, and discrete symmetries — with the primary derivation chain proved end-to-end.

The paper is organized as follows. §§2–4 establish the structural foundations: the minimal structure required by the theorems, the coupling graph interpretation of space, and the gauge freedom of the alphabet size. §§5–6 derive two geometric constraints: background independence and the unique dimension $d = 3$. §7 derives the emergent QFT from the lattice dynamics. §8 addresses the strong CP problem. §9 discusses the ontological and interpretive consequences, including relations to other discrete approaches. §10 synthesizes the complete program and proves the reconstruction theorem. §11 concludes.

1.1 Epistemic Status of the Results

The results fall into three tiers.

Tier 1 (structural foundations, from the six properties alone). The lattice is the coupling graph (§3), the alphabet size is gauge (§4), the observer is generic (§9.4), and $d = 3$ is the unique self-consistent dimension (§6). These are theorems about the equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$ and do not depend on which specific dynamics is selected.

Tier 2 (dynamics selection and the primary SM derivation chain). Center independence + isotropy + linearity uniquely select the wave equation (§7.1). The wave equation determines: staggered Dirac fermions, $K = 2d = 6$ from coupling-degree minimization, multiplicities (3, 2, 1) from the cubic group, local gauge invariance from background independence, chirality from the trace-out, unique hypercharges from anomaly cancellation, three generations from cubic symmetry, one Higgs doublet,

and $\theta = 0$ at all energy scales. Each step is proved; the chain is 29 theorems/lemmas long, and each depends on its predecessors. The individual proofs are short, but the cumulative chain is the paper's most exposed structure: a single failure at any link would propagate forward.

Tier 3 (redundant confirmations). Asymptotic freedom requiring $N_c \geq 3$ (Proposition), minimal chiral group (Proposition), and the Jordan-Chevalley projection (Appendix A). These provide independent second-route checks on Tier 2 results but are not load-bearing in the primary derivation.

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2. The Minimal Structure

2.1 What the theorems require

Each assumption in the companion paper [1] does specific work. The question is which assumptions are essential to the results and which are artifacts of the particular construction chosen.

Deterministic bijectivity. The P-indivisibility proof [1, §2.3] requires a bijection on a finite set: bijectivity guarantees recurrence ($\varphi^N = \mathrm{id}$), and recurrence produces the non-monotonic distinguishability that defines P-indivisibility. The entire framework rests on QM emerging from marginalizing deterministic dynamics — if the dynamics were already stochastic, there would be nothing to derive. Determinism is the core of the framework and cannot be weakened. However, determinism does not require a lattice. A continuous Hamiltonian system on a compact phase space is also deterministic and bijective (by Liouville's theorem). The lattice is one way to implement determinism on a finite set; it is not the only way.

Finite boundary entropy. The gap equation [1, §5] requires a finite number of degrees of freedom across the partition boundary: $S = A/\varepsilon^2$. This follows from the holographic entropy bound [3] applied to any finite-area surface, not from lattice structure. What matters is that the boundary carries finitely many modes — not that the bulk is a lattice.

Bounded coupling degree (locality). This assumption does the most work. The area law (§5) follows from the spatial Markov property, which requires range-1 coupling. Without bounded coupling, long-range correlations could produce volume-law entropy, breaking the Jacobson route to GR. Locality does not require a regular lattice. Any bounded-degree graph has the spatial Markov property for range-1 dynamics. The theorems would work on a random graph with bounded degree, as long as it has the right statistical properties.

Statistical homogeneity and isotropy. The Myrheim-Meyer dimension [4] requires a well-defined notion of spacetime dimension, which emerges from the causal order statistics on a homogeneous, isotropic structure. The dispersion relation (§5) uses translation invariance. The dynamics selection (§7.1) uses isotropy. Exact lattice regularity is sufficient but not necessary. A random graph with statistical homogeneity and isotropy at large scales would produce the same results.

Non-trivial coupling (C1) and slow bath with capacity (C2, C3). These are conditions on the partition, not on the lattice. They hold for any system with the right partition geometry, regardless of whether the underlying space is a lattice, a graph, or something else.

2.2 What the theorems do not require

A *regular lattice*. Any bounded-degree graph with statistical isotropy suffices. A *specific alphabet size q*. All proofs work for any q ≥ 2; no prediction depends on q. A *specific dimensionality d*. The proofs work for any d ≥ 1; self-consistency selects d = 3 (§6). The *wave equation*. It is derived from center independence, isotropy, and linearity (§7.1), each of which follows from the structural properties listed above.

2.3 The minimal object

The theorems require exactly: a deterministic bijection on a finite set, whose state space factors into local degrees of freedom coupled by a bounded-degree graph with statistical isotropy, partitioned into visible and hidden sectors satisfying C1–C3. Everything else — the regular lattice, the alphabet, the dimensionality, the wave equation, quantum mechanics, general relativity — is either derived or irrelevant. The minimal object is the triple (S, φ, V): a finite set S, a bijection φ: S → S with bounded-degree coupling, and a partition V ⊂ S of the degrees of freedom into visible and hidden sectors.

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3. The Coupling Graph

3.1 Definition and the factorization problem

Let $S = S_1 \times S_2 \times \ldots \times S_N$ be a product of finite sets (the local state spaces), and let φ: S → S be a bijection. The *coupling graph* G_φ is the undirected graph on vertices {1, …, N} where i and j are connected if the i-th component of φ(s) depends on the j-th component of s for at least one state s ∈ S. The coupling graph is not an additional structure imposed on the system — it is read off from φ.

What determines the product decomposition? An abstract finite set S does not come with a canonical factorization. The answer is that the factorization is determined by φ itself, as the one that *minimizes the coupling degree*. For the wave equation, numerical verification confirms this: the natural spatial factorization is the *unique minimizer* of coupling degree — 0% of random factorizations achieve it, and 100% have strictly higher coupling.

Theorem (factorization uniqueness for translation-invariant dynamics). *Let φ be a bijection on $S = (\mathbb{Z}/q\mathbb{Z})^N$ satisfying: (a) translation invariance, (b) range-1 coupling on the natural factorization, and (c) genuinely bidirectional coupling. Then the natural factorization is the unique minimizer of coupling degree among all product decompositions $S = S_1 \times \cdots \times S_M$ with $|S_k| = q$ for all k.* The key step is that

the wave equation's addition $x_{i-1} + x_{i+1} - x_i^{\text{prev}}$ creates irreducible algebraic dependence among all input components (for $q > 2$), so any relabeling that splits or merges natural sites increases the coupling degree.

Extension to general graphs. The translation-invariant proof extends to any bounded-degree graph with polynomial growth exponent d and statistical isotropy. The argument proceeds in three steps. First, §6 establishes (via Gromov's theorem) that any such graph is quasi-isometric to $\mathbb{Z}^d$, so the asymptotic dimension is d. Second, the wave equation on a d-dimensional graph has $2d$ independent low-energy propagation modes (from heat kernel asymptotics: the spectral dimension equals d, giving d momentum-like parameters over two orientations each). Third, the integer-valued coupling degree $\delta(K)$ depends on the graph only through its spectral properties (heat kernel decay exponent, spectral dimension), which vary continuously under quasi-isometric deformations; since K is integer-valued and the minimizer is $K = 2d$ for $\mathbb{Z}^d$, it cannot change without a discontinuous jump in the spectral dimension — which quasi-isometries preserve. The factorization uniqueness is therefore a property of the equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$, not of the specific graph.

3.2 Space as coupling structure

Every "spatial" property used in the derivation chain is a property of G_φ. The *area* of a region V is the number of edges crossing from V to its complement. The *dimension* is extracted from the causal partial order via the Myrheim-Meyer estimator [4]. The *area law* follows from the spatial Markov property on any graph with range-1 dynamics. The *dispersion relation* requires statistical isotropy but not exact regularity. The regular cubic lattice is one graph with these properties; it is not the only one.

3.3 The dissolution of the container problem

The coupling graph is not embedded in a pre-existing space. There is no ambient manifold in which the graph "lives." The vertices are labels for degrees of freedom; the edges record which degrees of freedom are dynamically coupled. Any spatial embedding is a representation for human convenience, not a physical fact. The coupling graph is not made of material. The vertices are not atoms of space, and the edges are not physical connections. A degree of freedom is a component of the state — it is an abstract mathematical entity, not an object. Asking "what is site i made of?" is a category error: it is a label for a variable, not a name for a thing.

The most common objection to discrete models of spacetime is the container problem: if space is a lattice, what does the lattice sit in? On the structural reading, the container problem does not arise. The coupling graph is defined by the bijection φ, not by embedding in a manifold. Two sites are neighbors because φ couples them — because the state at one affects the future of the other. This is a dynamical fact, not a spatial one. Locality is defined by the dynamics, and space emerges from locality. At no point does anything need to be "inside" anything else. The relationship is analogous to a social network: two people are "connected" not because they occupy adjacent points in physical space, but because they interact. The topology is defined by the interactions, not by geography.

4. The Alphabet as Gauge Freedom

Every prediction of the OI framework is independent of the alphabet size q. The gap equation $\hbar = c^3\varepsilon^2/(4G)$ contains no q. The Bekenstein-Hawking formula contains no q. The dispersion relation, area law, P-indivisibility, center independence, and Einstein's equations are all q-independent. Numerical confirmation: on a 1D ring of L = 40 sites, the wave equation produces significant P-indivisibility at every q tested from 2 to 64. No experiment, even in principle, could measure q.

Two systems (S, φ) and (S', φ') with different alphabet sizes q and q' are *physically equivalent* if they produce the same emergent transition probabilities. The analogy to electromagnetic gauge invariance is precise: the gauge potential A_μ is not physical; the field strength $F_{\mu\nu}$ is. In the OI framework, the mod-q state space is not physical; the coupling structure and its emergent consequences are.

5. Background Independence

The companion paper [1] treats the coupling graph as fixed. In general relativity, the spacetime geometry is dynamical. If space is the coupling graph, background independence requires the graph to evolve with the state: $s(t+1) = \varphi_{s(t)}(s(t))$, where each φ_s is a bijection but G_{φ_s} varies with s.

Bijectivity is automatic. The wave equation is second-order: $x_i(t+1) = f(\text{neighbors of i at time t}) - x_i(t-1) \bmod q$. The phase-space map $F: (x(t-1), x(t)) \to (x(t), x(t+1))$ has an explicit inverse that uses x(t) to determine which graph to apply. Bijectivity of the phase-space map is automatic for any second-order reversible dynamics, regardless of whether the coupling is state-dependent. The P-indivisibility proof applies without modification.

The derivation chain survives under three constraints. (i) *Local graph-dependence:* G(x) at site i depends only on x_j within a bounded range of i. (ii) *Center-independent graph-dependence:* G(x) at site i does not depend on x_i, preserving center independence. (iii) *Statistical isotropy:* G(x) is statistically isotropic on large scales for typical configurations.

Under these constraints: the area law holds, center independence is preserved, the gap equation is unchanged, and the dynamics selection theorem applies at each step.

Explicit construction. On a ring of L ≥ 4 sites with alphabet $\mathbb{Z}/q\mathbb{Z}$, define the right-neighbor assignment $R(i,x) = i + 2 \bmod L$ if $x_{(i+1) \bmod L} = 0$, and $R(i,x) = i + 1 \bmod L$ otherwise, with left neighbor $L(i) = i - 1 \bmod L$ fixed. The dynamics is $x_i(t+1) = (x_{L(i)}(t) + x_{R(i,x(t))}(t) - x_i(t-1)) \bmod q$. This satisfies: constraint (i) with range 1 ($R(i,x)$ depends only on x_{i+1}); constraint (ii) exactly ($R(i,x)$ does not depend on x_i); constraint (iii) by translation invariance. Bijectivity is verified by explicit inverse: $u_i = (v_{L(i)} + v_{R(i,v)} - w_i) \bmod q$, since v determines all neighbor

assignments. Center independence, P-indivisibility, and graph regularity (undirected degree bounded in [2, 4] independently of the state) all hold for any L and q.

Local gauge invariance from background independence. With state-dependent coupling, the coupling matrix M (§7.4 below) becomes site-dependent: $M = M(\mathbf{n}, \hat{e}_j)$. At every site, $M(\mathbf{n}, \hat{e}_j)$ has K = 6 components with eigenvalue multiplicities (3, 2, 1). Since M varies from site to site, the commutant $\mathrm{U}(3) \times \mathrm{U}(2) \times \mathrm{U}(1)$ acts independently at each site. Under a site-dependent transformation $G(\mathbf{n})$:

$$\phi(\mathbf{n}) \to G(\mathbf{n})\, \phi(\mathbf{n}), \qquad M(\mathbf{n}, \hat{e}_j) \to G(\mathbf{n})\, M(\mathbf{n}, \hat{e}_j)\, G(\mathbf{n} + \hat{e}_j)^{-1}$$

The wave equation is invariant. This is local gauge invariance. The link variable $M(\mathbf{n}, \hat{e}_j)$ transforms as a gauge connection: the plaquette product $P(\mathbf{n}, j, k) = M(\mathbf{n}, \hat{e}_j)\, M(\mathbf{n} + \hat{e}_j, \hat{e}_k)\, M(\mathbf{n} + \hat{e}_k, \hat{e}_j)^{-1}\, M(\mathbf{n}, \hat{e}_k)^{-1}$ transforms in the adjoint ($P \to G(\mathbf{n})\, P\, G(\mathbf{n})^{-1}$), so $\mathrm{Re}\,\mathrm{Tr}(P)$ is gauge-invariant — it is the Wilson plaquette action [9], now derived rather than postulated.

Einstein's equations as self-consistency. The Jacobson thermodynamic argument [24], applied to local causal horizons on the state-dependent graph, requires three inputs — each proved here for the wave equation.

Lemma (Area law). *The entanglement entropy of a spatial region V in the wave equation scales as $S(V) = \eta\, |\partial V|$. This holds for both the linear wave equation over $\mathbb{R}$ and the mod-q wave equation over $\mathbb{Z}/q\mathbb{Z}$.*

Proof. Over $\mathbb{R}$: the wave equation is a system of coupled harmonic oscillators with a quadratic Hamiltonian determined by the adjacency matrix. The area-law theorem for Gaussian systems applies directly [2]. Over $\mathbb{Z}/q\mathbb{Z}$: the wave equation has range $R = 1$, so for any $i \in V^\circ$ (interior of V), all neighbors lie inside V. With uniform initial conditions: $V^\circ \perp V^c \mid \partial V$ (spatial Markov property). By the chain rule: $I(V; V^c) = I(\partial V; V^c) \leq H(\partial V) \leq |\partial V| \cdot \log_2 q$. The mutual information scales as boundary area, not volume. □

Lemma (Relativistic dispersion). *The wave equation has exact dispersion relation* $\cos\omega = \cos k$.

Proof. Substituting $x_j(t) = A \exp(i(kj - \omega t))$: $e^{-i\omega} + e^{i\omega} = e^{-ik} + e^{ik}$, giving $\cos\omega = \cos k$. For small k: $\omega = |k| + O(k^3)$ — relativistic propagation with $v = 1$. □

Lemma (Lattice Bisognano-Wichmann). *For the wave equation, whose low-energy effective theory is Lorentz-invariant, the entanglement Hamiltonian for a half-space bipartition has the Bisognano-Wichmann form. The reduced state for a Rindler observer is thermal at $T_U = \hbar\kappa/(2\pi c k_B)$, with corrections at $O(\varepsilon^2\kappa^2/c^2)$.*

Proof. The mod-q wave equation is a coupled harmonic oscillator system — the class for which the lattice BW theorem is proved analytically [23]. Its low-energy dispersion is relativistic (the preceding lemma). Thermality of the Unruh effect is robust against UV modifications of the dispersion relation, with corrections scaling as $O(\varepsilon^2\kappa^2/c^2)$. □

Theorem (Discrete Einstein equation). *For the state-dependent wave equation on a bounded-degree graph G(x) satisfying constraints (i)–(iii), the Jacobson thermodynamic argument produces:*

$$\kappa_{OR}(i, j;\, G(x)) = \alpha \cdot \mathcal{T}_{ij}(x,\, G(x))$$

where κ_{OR} is the Ollivier-Ricci curvature [21], $\mathcal{T}_{ij}$ is the discrete stress-energy, and $\alpha = 8\pi G/(c^4\eta)$.

Proof. (i) *Area law.* The entanglement entropy scales as $S(V) = \eta\,|\partial V|$ (area-law lemma above), holding for any bounded-degree graph with range-1 dynamics. (ii) *Unruh temperature.* The lattice Bisognano-Wichmann lemma gives $T = \hbar\kappa/(2\pi c k_B)$ at local causal horizons. (iii) *Clausius relation.* At each edge (i, j): the area variation under graph deformation is captured by κ_{OR} (which measures, via optimal transport, the deviation from flat space), giving $\delta A/A = -\kappa_{OR}\,\delta\lambda$. Substituting into $\delta Q = T\,\delta S$: $\mathcal{T}_{ij}\,\delta\lambda\,\delta A = T \cdot \eta \cdot (-\kappa_{OR}\,\delta\lambda\,A)$, which rearranges to $\kappa_{OR} = \alpha \cdot \mathcal{T}_{ij}$. (iv) *Continuum limit.* Van der Hoorn et al. [15] prove $\kappa_{OR} \to$ Ric on random geometric graphs converging to a Riemannian manifold. Kelly et al. [16] prove $\sum \kappa_{OR} \to \int R\,\sqrt{g}\,d^d x$. Together: the discrete Einstein equation converges to $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)\,T_{\mu\nu}$. □

Every orbit is self-consistent by construction: the state determines the graph, which determines the entropy and temperature, and the Clausius relation constrains the graph's response to energy flux — paralleling continuum GR where the Einstein equations are the equations of motion, not constraints on separate dynamics. The mathematical framework is unchanged: the full dynamics F(u, v) = (v, WE_{G(v)}(v) − u) is a single bijection on the finite phase space. The ontological object is still (S, F, V).

Lattice corrections to Einstein's equations. The derivation invokes continuum results in the final step (Jacobson). Decomposing Jacobson's argument into its four inputs and checking each independently: (i) $\delta S = \eta\,\delta A/\varepsilon^2$ (area-law entropy) — proved for Gaussian systems over $\mathbb{R}$ [2] and for any nearest-neighbor dynamics over $\mathbb{Z}/q\mathbb{Z}$ via the spatial Markov property (area-law lemma above). No Lorentz invariance enters. **Exact on the lattice.** (ii) $T = \hbar\kappa/(2\pi c k_B)$ (Unruh temperature) — requires the Bisognano-Wichmann theorem. On the lattice, the BW form is proved analytically for coupled harmonic oscillator systems [23] and shown robust against UV modifications of the dispersion relation with corrections at $O(\varepsilon^2\kappa^2/c^2)$. **Approximate.** (iii) $\delta Q = \int T_{\mu\nu} k^\mu d\Sigma^\nu$ (energy flux) — a definition on the emergent manifold. No Lorentz invariance required. **Exact.** (iv) The Raychaudhuri equation — kinematic: it holds on any pseudo-Riemannian manifold as a consequence of the definition of curvature. **Exact.** Since inputs (i), (iii), and (iv) are lattice-exact, corrections enter only through input (ii):

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \times \left[1 + \mathcal{O}\left(\frac{\varepsilon^2\kappa^2}{c^2}\right)\right]$$

For the cosmological horizon, $\varepsilon\kappa/c = 2l_P \cdot H/c \sim 10^{-61}$, giving corrections of order

10^{-122}. For solar-mass black holes, $\varepsilon\kappa/c \sim 10^{-38}$, giving corrections of order 10^{-76}. The corrections are negligible for any horizon much larger than the Planck scale.

The $\eta = 1/4$ coefficient. The Bekenstein-Hawking formula $S = A/(4l_P^2)$ follows from $\varepsilon = 2l_P$ (the gap equation [1, §5]): one entropy unit per cell of area $\varepsilon^2 = 4l_P^2$, giving $S = A/\varepsilon^2 = A/(4l_P^2)$. This cannot be confirmed via entanglement entropy because of the species problem — the entanglement entropy coefficient is species-dependent (Srednicki found $\eta_{\text{ent}} \approx 0.295$ per scalar field), while the BH entropy is species-independent. The thermal matching route avoids this by counting classical boundary degrees of freedom (one per ε^2 cell, independent of field content) rather than any particular field's vacuum entanglement.

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6. Why d = 3

The theorems work for any d ≥ 1. Four independent self-consistency filters converge on d = 3.

The concordance calculation. In d spatial dimensions, the boundary entropy ratio is:

$$\frac{\rho_s}{\rho_{\text{crit}}} = \frac{2}{d-1}$$

This equals unity only for d = 3. For d = 2: overclosure. For d ≥ 4: gravitational deficit with no source.

d	ρ_s / ρ_crit	Status
2	2	Overclosure
3	**1**	**Exact concordance**
4	2/3	Deficit

Propagating gravity requires d ≥ 3. The Weyl tensor vanishes for d ≤ 2. GW echoes and the Jacobson derivation require propagating gravitational degrees of freedom.

Stable matter requires d ≤ 3. The Coulomb potential in d dimensions gives unstable atoms for d ≥ 4 (Ehrenfest [13]). Gravitational orbits are unstable for d ≥ 4 [14]. Without stable matter, no observers exist to define the partition.

Renormalizability requires d = 3. The Yang-Mills coupling has mass dimension [g] = (3−d)/2; dimensionless only for d = 3.

The intersection of d ≥ 3, d ≤ 3, the concordance, and renormalizability is d = 3 uniquely.

Integer dimension is derived, not presupposed. The self-consistency argument might appear to presuppose that the coupling graph has a well-defined integer dimension. In fact, every bounded-degree graph falls into one of three growth classes: expo-

nential ($|B(r)| \sim k^r$), fractal ($|B(r)| \sim r^\alpha$ with non-integer α), or integer-polynomial ($|B(r)| \sim r^d$ with integer d). Self-consistency excludes the first two.

Exponential growth fails because the wave equation on such graphs does not produce Lorentz-invariant dispersion (the spectral gap doesn't close correctly), so Jacobson's derivation of Einstein's equations fails. Additionally, $d_{\text{eff}} \to \infty$ is incompatible with stable matter ($d \leq 3$) and renormalizability ($d = 3$).

Fractal growth fails because Jacobson's argument requires the Raychaudhuri equation, which requires geodesic congruences on a manifold — fractal spaces don't support them. The staggered fermion decomposition requires a hypercubic BZ structure ($\eta \in \{0, 1\}^d$), which doesn't exist on fractal graphs. The cubic group decomposition of $2d$ link directions — which constrains the gauge structure (§7.6) — requires cubic lattice symmetry, absent in fractals.

Only *integer-polynomial growth* survives. By Gromov's theorem (finitely generated groups of polynomial growth are virtually nilpotent) and statistical isotropy, the graph is quasi-isometric to $\mathbb{Z}^d$ for integer d. The four filters then select $d = 3$.

Corollary ($d = 3$ is necessary, not contingent). *Any bijection whose coupling graph has $d \neq 3$ produces internal contradictions in its emergent description. The coupling graph dimension is not a free parameter or a property of a specific φ — it is the unique value compatible with self-consistency.*

Relation to the anthropic principle. This is not a standard anthropic argument. The standard argument invokes a multiverse of dimensions and selection bias: many d exist; we observe $d = 3$ because only there can observers exist. The present argument is different: the framework's axioms require an observer as a constitutive element (Axiom 3 defines the partition relative to an observer). The requirement that such an observer exist, combined with the framework-internal concordance calculation, constrains d. The observer is not selecting from a pre-existing landscape but is a structural prerequisite for the axioms to have content.

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7. The Emergent Quantum Field Theory

The wave equation on a d = 3 lattice determines not just QM and GR but the specific QFT the observer sees. This section derives the Standard Model's structure through three complementary routes: bottom-up from the lattice dynamics to fermionic matter (§§7.1–7.3), gauge emergence from multi-component dynamics (§§7.4–7.6), and the staggered species / consistency constraints that fix the matter content and discrete symmetries (§§7.7–7.12).

7.1 The wave equation as lattice Klein-Gordon

The OI lattice is a d-dimensional hypercubic lattice Λ = ℤ^d with spacing ε = 2 l_p. The dynamics on this lattice is not a free choice — it is uniquely determined by three requirements.

Theorem (Dynamics selection). *Among all second-order reversible nearest-neighbor dynamics on a d-dimensional lattice of alphabet size q, the requirements of (i) center independence, (ii) spatial isotropy, and (iii) linearity uniquely select the discrete wave equation. This holds for any q ≥ 2 and d ≥ 1.*

Proof. The update function has the form $x_i(t+1) = (f(\text{neighbors of } i \text{ at time } t) - x_i(t-1)) \bmod q$. Center independence requires that f not depend on x_i itself: f reduces to a function of the $2d$ neighboring values only. Spatial isotropy requires f to be symmetric under all permutations of the neighbors corresponding to lattice symmetries. Linearity over $\mathbb{Z}/q\mathbb{Z}$ requires $f(a+a') = f(a) + f(a')$. The unique function satisfying all three is $f = \alpha(x_1 + x_2 + \cdots + x_{2d}) \bmod q$ for some constant α. The propagation speed is $v = \alpha$, so $\alpha = 1$ gives the maximum lattice-scale speed. This is the discrete wave equation. □

Each requirement has a physical justification.

Center independence is necessary for emergent QM. The mechanism is information screening. For the checkerboard partition, the adjacency matrix A is bipartite: $A_{VV} = 0$ (no visible–visible edges). In the center-independent (CI) case, the update $x_i(t+1) = [h_{i-1}(t) + h_{i+1}(t) - x_i(t-1)] \bmod q$ does not contain $x_i(t)$. Therefore $X_V(t+1)$ carries no information about $X_V(t)$; conditioning on $X_V(t+1)$ constrains $H(t)$ but does not screen $X_V(t)$ from the future. Since $H(t+1)$ depends on $X_V(t)$, the path $X_V(t) \to H(t+1) \to X_V(t+2)$ bypasses the present: the process is non-Markovian. In the center-dependent (CD) case, the update $x_i(t+1) = [x_i(t) + h_{i-1}(t) + h_{i+1}(t) - x_i(t-1)] \bmod q$ contains $x_i(t)$ explicitly. Conditioning on $(X_V(t), X_V(t+1))$ gives $|V|$ linear equations in $|H|$ unknowns over $\mathbb{Z}/q\mathbb{Z}$; generically (for q prime) the hidden state $H(t)$ is fully determined, so the augmented process is Markov. Center coupling makes the present an explicit function of the visible past, allowing it to determine the hidden state and screen the past from the future. Without center coupling, the hidden sector retains unscreened correlations — which is P-indivisibility.

Spatial isotropy is required for Lorentz invariance: without rotational symmetry, the dynamics has a preferred spatial direction, breaking the isotropy that Lorentz invariance demands.

Linearity is selected by four independent criteria. First, q-gauge invariance (§4): the alphabet size q is proved gauge — no prediction depends on it. Linear dynamics over $\mathbb{Z}/q\mathbb{Z}$ has q-independent structural properties (transition probabilities, dispersion, P-indivisibility); non-linear dynamics generically does not ($x^2 \bmod 3$ and $x^2 \bmod 5$ have different algebraic structure), so the emergent physics would depend on q, contradicting the gauge freedom. Linearity is necessary, not merely selected. Second, over $\mathbb{R}$, nonlinear center-independent dynamics have propagation speed $v < 1$, so linearity gives maximum speed. Third, over $\mathbb{Z}/q\mathbb{Z}$, among all linear CI dynamics $f = \alpha(l + r) \bmod q$ with prime $q \geq 5$, $\alpha = 1$ uniquely maximizes P-indivisibility. Fourth, modified (nonlinear) dispersion relations break the thermality of the Unruh effect — the KMS condition holds only for linear dispersion $\omega = |k|$. Since the Jacobson derivation (§5) requires thermal equilibrium at horizons, only the wave equation supports the full GR chain. All four criteria converge on the same dynamics.

$$\phi(n,t+1) = \phi(n-1,t) + \phi(n+1,t) - \phi(n,t-1)$$

Theorem 1. *The OI wave equation is the massless lattice Klein-Gordon equation. Center independence is equivalent to the vanishing of the lattice mass term.*

Proof. The general linear update is $\phi(n,t+1) = \alpha\,\phi(n,t) + \beta[\phi(n-1,t) + \phi(n+1,t)] + \gamma\,\phi(n,t-1)$. Reversibility: $\gamma = -1$. Center independence: $\alpha = 0$. Isotropy + unit speed: $\beta = 1$. The d'Alembertian $\Box_{\text{lat}}\phi = -\alpha\,\phi = 0$: massless KG. □

7.2 Factorization into Dirac operators

Theorem 2 (Susskind [5]). $D_{st}^2 = -\frac{1}{4}\Box_{lat}$. *The OI wave equation is equivalent to the massless staggered Dirac equation.*

The bosonic and fermionic descriptions are related by an exact change of variables on the lattice.

7.3 Center independence and chiral symmetry

Theorem 3. *Center independence of the lattice dynamics is equivalent to exact chiral symmetry of the emergent staggered fermions.*

Proof. The staggered mass term squares to $\Box_{\text{lat}}\phi = -4m_{\text{lat}}^2\phi$: center dependence. Conversely, center independence gives $D_{\text{st}}\chi = 0$, invariant under $\chi \to \varepsilon\chi$ (chiral symmetry). □

The chain: P-indivisibility → center independence → chiral symmetry → Higgs mechanism [6, 7]. One algebraic condition ($\alpha = 0$) produces quantum mechanics, chiral fermions, and the Higgs boson.

7.4 Multi-component dynamics and gauge structure

Each site now carries a K-component vector $\phi(\mathbf{n},t) \in (\mathbb{Z}/q\mathbb{Z})^K$. The general second-order linear update is $\phi(\mathbf{n},t+1) = C\,\phi(\mathbf{n},t) + \sum_j M^{(j)}[\phi(\mathbf{n}+\hat{e}_j,t) + \phi(\mathbf{n}-\hat{e}_j,t)] + D\,\phi(\mathbf{n},t-1)$, where $C, D, M^{(j)} \in \text{Mat}(K)$. Reversibility requires $D = -I_K$; center independence requires $C = 0$; spatial isotropy requires $M^{(j)} = M$ for all j. This uniquely selects the matrix wave equation:

$$\phi(\mathbf{n},t+1) = M\sum_{j=1}^{d}\left[\phi(\mathbf{n}+\hat{e}_j,t) + \phi(\mathbf{n}-\hat{e}_j,t)\right] - \phi(\mathbf{n},t-1)$$

where $M \in \text{Mat}(K)$ is the sole free parameter.

Theorem 4 (Mass spectrum). *Each eigenvalue μ_a of M determines a dispersion relation $\cos\omega_a = \mu_a \cos k$. Massless iff $\mu_a = 1$. Stability requires $|\mu_a| \le 1$.*

Proof. Diagonalizing $M = V\,\text{diag}(\mu_1,\dots,\mu_K)\,V^{-1}$, each eigenmode decouples. Substituting the plane wave $\phi_a \propto e^{i(kn-\omega t)}$ into the a-th decoupled equation gives $e^{-i\omega} +$

$e^{i\omega} = \mu_a(e^{-ik} + e^{ik})$, hence $\cos\omega_a = \mu_a \cos k$. For $|\mu_a| > 1$, there exist real k with $|\mu_a \cos k| > 1$, giving complex ω (exponential growth). □

Theorem 5 (Gauge group). *The global gauge group is the commutant $G = C_{\mathrm{U}(K)}(M)$. If M has r distinct eigenvalues with multiplicities $(n_1, \dots, n_r)$: $G = \mathrm{U}(n_1) \times \cdots \times \mathrm{U}(n_r)$.*

Proof. A global transformation $\phi \to U\phi$ with $U \in \mathrm{U}(K)$ preserves the matrix wave equation iff $UMU^{-1} = M$, i.e., $[U, M] = 0$. By Schur's lemma, the set of unitaries commuting with M decomposes as a product of unitary groups, one on each eigenspace. □

The gauge group and mass spectrum are the same information: modes of equal mass share a gauge symmetry.

7.5 K = 2d from coupling-degree minimization

Theorem 6 ($K = 2d$). *For any bounded-degree graph with polynomial growth exponent d and statistical isotropy, the wave equation (selected by center independence, isotropy, and linearity) has a unique minimal-coupling factorization with $K = 2d$ components.*

Proof (translation-invariant case). The matrix wave equation on $\mathbb{Z}^d$ updates the K-component vector at site $\mathbf{n}$ using exactly 2d independent spatial inputs: $\{\phi(\mathbf{n} \pm \hat{e}_j, t)\}_{j=1}^{d}$. Define the *internal coupling degree* δ(K) as the maximum number of spatial input channels on which any single component's update depends.

Step 1 (K = 2d achieves δ = 1). With K = 2d, assign component a to link direction $a \in \{\pm\hat{e}_1, \dots, \pm\hat{e}_d\}$. If M is diagonal in this basis, then $\phi_a(\mathbf{n}, t+1) = \mu_a \cdot \phi_a(\mathbf{n} + \hat{e}_{j(a)}, t) - \phi_a(\mathbf{n}, t-1)$: each component depends on exactly one spatial input. Therefore δ(2d) = 1.

Step 2 (K < 2d forces δ ≥ 2). By the pigeonhole principle, at least one component must receive contributions from $\lceil 2d/K \rceil \geq 2$ spatial input channels. These channels carry independent data from distinct neighboring sites, so δ(K) ≥ 2.

Step 3 (K > 2d forces δ ≥ 2). With K > 2d components and only 2d spatial input channels, at least K − 2d > 0 components have no dedicated spatial input. The matrix M has rank at most 2d in its spatial coupling structure (since it acts on 2d independent inputs), so the K × K coupling necessarily maps excess components to linear combinations of other components' spatial inputs. These excess components therefore depend on at least 2 spatial channels through M: δ(K) ≥ 2.

Uniqueness. Steps 1–3 show δ = 1 iff K = 2d.

Extension to general graphs. For a bounded-degree graph G with polynomial growth exponent d and statistical isotropy: (i) G is quasi-isometric to $\mathbb{Z}^d$ (§6, Gromov's theorem); (ii) the wave equation on G has $2d$ independent low-energy propagation modes, determined by the spectral dimension $d_s = d$ (from heat kernel asymptotics $p_t(x, x) \sim t^{-d/2}$ on graphs with polynomial growth exponent d); (iii) the coupling-degree minimizer $K = 2d$ on $\mathbb{Z}^d$ is stable under quasi-isometric deformations,

because $\delta(K)$ depends on the graph only through its spectral dimension, which quasi-isometries preserve — and K is integer-valued, so it cannot change without a discontinuous jump in d_s. □

This is the per-site analog of the global factorization principle (§3.1): at the global level, $S = S_1 \times \cdots \times S_N$ is the unique factorization minimizing coupling degree — defining "space"; at the per-site level, K components minimize internal coupling degree — defining "gauge structure." The objection that "nature need not minimize coupling degree" mistakes the logical role. The framework does not postulate a preference for minimal coupling — coupling-degree minimization is the mathematical definition of "site" given a bijection, and the factorization is unique (§3.1). A system with $K = 8$ or $K = 4$ would correspond to a *different bijection* φ', not to the same φ with a non-minimal factorization. Since the dynamics is selected by center independence, isotropy, and linearity (§7.1), and its factorization is unique, $K = 2d = 6$ is a theorem about the wave equation on $\mathbb{Z}^3$, not a selection rule.

7.6 Cubic decomposition: multiplicities (3, 2, 1)

Theorem 7. *The 2d = 6 nearest-neighbor link directions of the d = 3 cubic lattice decompose under the rotation group O as* $6 = T_1(3) \oplus E(2) \oplus A_1(1)$. *The eigenvalue multiplicities of M are (3, 2, 1), giving gauge group* $\mathrm{U}(3) \times \mathrm{U}(2) \times \mathrm{U}(1) \supset \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$.

Proof. Characters at each conjugacy class of O (24 elements): E: $\chi = 6$; $8C_3$: $\chi = 0$; $3C_2$: $\chi = 2$; $6C_2'$: $\chi = 0$; $6C_4$: $\chi = 2$. Character inner products give $n(A_1) = 1$, $n(E) = 1$, $n(T_1) = 1$, with $n(A_2) = n(T_2) = 0$. By Schur's lemma and isotropy, $M = \mathrm{diag}(\mu_c I_3, \mu_w I_2, \mu_y)$. □

The max-speed constraint requires $\mu_y = 1$ (the A_1 mode propagates at the lattice speed of light). The physical identifications: T_1 (spatial vector) → color, E (quadrupole) → weak isospin, A_1 (scalar) → hypercharge. Local gauge invariance follows from background independence (§5).

Remark (robustness under statistical isotropy). Theorem 7 uses the character table of the octahedral group O, the exact symmetry of the cubic lattice. The structural reading of §2 claims that any bounded-degree graph with statistical isotropy suffices. The robustness follows from Theorem 6's general proof and two additional observations. First, §6 establishes that the coupling graph is quasi-isometric to $\mathbb{Z}^3$ (Gromov's theorem + statistical isotropy + integer polynomial growth). The relevant representation theory is that of the large-scale rotation group $O(3)$; the decomposition of the 2d = 6 link directions into irreducible representations has multiplicities $(3, 2, 1)$ determined by the topology of $d = 3$ rotations, not by the precision of the lattice symmetry. Second, the gauge group is a discrete structure: the eigenvalue multiplicities of M are integers, and Theorem 5 maps integer multiplicities to unitary group factors. Small perturbations of the graph — breaking exact O symmetry to statistical isotropy — perturb the eigenvalues of M continuously but cannot change integer multiplicities without crossing a codimension-1 boundary in parameter space. Both $K = 2d$ (Theorem 6) and the $(3, 2, 1)$ decomposition are therefore topologically stable under all deformations within the equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$.

7.7 Staggered species, generations, and the Higgs

The Nielsen-Ninomiya theorem [8] requires 2^{d+1} Weyl species. Staggered reduction gives $N_{\text{taste}} = 2^{\lfloor (d+1)/2 \rfloor} = 4$ Dirac tastes in d+1 = 4.

Theorem 8. *Under the cubic group O:* $4 = \mathbf{1} \oplus \mathbf{3}$*. One singlet taste and three triplet tastes.*

Proof. The 2^d = 8 BZ corners pair into 4 taste pairs under $\eta \leftrightarrow \mathbf{1} - \eta$. The pair {(0,0,0), (1,1,1)} is O-invariant (singlet A_1). The three axis-aligned pairs form irreducible T_1. □

Theorem 8b (Taste coupling structure). *On the d = 3 lattice with the normalized wave equation, each BZ corner* η *has coupling* $\mu(\eta) = \sigma(\eta)/d$ *where* $\sigma = \sum_j \cos(\pi\eta_j)$*:*

$$\mu_\Gamma = 1, \quad \mu_X = \tfrac{1}{3}, \quad \mu_M = -\tfrac{1}{3}, \quad \mu_R = -1$$

The staggered taste pairs have couplings: singlet (Γ*,* R*):* $|\mu| = 1$*; triplet (*X_j*,* M_j*):* $|\mu| = 1/3$*, identical for all* $j = 1, 2, 3$*.*

Proof. Direct computation: $\sigma(\Gamma) = 3$, $\sigma(X) = 1$, $\sigma(M) = -1$, $\sigma(R) = -3$. Dividing by $d = 3$ gives the couplings. The triplet members are related by cubic symmetry (O permutes spatial axes), hence identical. □

Theorem 9 (Spin-taste correspondence). *(a) The singlet taste produces a spin-0 field. (b) The triplet taste produces spin-1/2 fields.*

Proof. (a) The singlet pairs $\Gamma(0,0,0) = I_4$ (scalar) and $\Gamma(1,1,1) = \gamma^1\gamma^2\gamma^3$ (pseudoscalar), both SO(3)-invariant: J = 0. (b) The j-th triplet pairs γ^j (vector) with $-i\Sigma_j$ (spin generator): J = 1/2. □

Corollary (Spin-statistics from the lattice). Singlet → spin-0 → boson (Higgs). Triplet → spin-1/2 → fermion (quarks and leptons).

Theorem 10 (Three Standard Model generations). *The three triplet staggered tastes, each producing a spin-1/2 Dirac fermion with K = 6 internal components, constitute three complete SM generations with the anomaly-cancelling representations.*

Proof. (i) Three spin-1/2 sectors and one spin-0 sector (Theorem 9). (ii) Anomaly cancellation (Theorem 15 below) uniquely determines the matter representations per generation. No alternative anomaly-free assignment exists [10]. (iii) The three sectors are related by cubic symmetry (O permutes spatial axes). (iv) Three degenerate sectors, each with unique anomaly-cancelling representations, constitute three complete SM generations. □

Theorem 11 (Higgs sector). *The singlet taste produces spin-0 excitations with quantum numbers* $(\mathbf{1}, \mathbf{2}, +\frac{1}{2})$ *— exactly the Higgs doublet.*

Proof. The singlet taste is spin-0 (Theorem 9a). The tensor product with the E(2) sector gives $A_1 \otimes E = E$: the Higgs couples to SU(2) as a doublet. The hypercharge +1/2 is

uniquely determined by gauge-invariant Yukawa couplings and anomaly cancellation. □

Consequences. Exactly one Higgs doublet (one singlet taste). No ν_R: the singlet taste produces the Higgs, not a fourth generation. Neutrinos are Majorana (Weinberg operator).

Theorem 12 (Partition = spinor decomposition). *The OI checkerboard partition coincides with the staggered spinor decomposition: visible and hidden sectors carry complementary spinor components.*

Proof. The checkerboard assigns site **n** to the visible sector iff $(-1)^{\sum n_\mu} = +1$ (even sublattice). In the staggered-to-Dirac reconstruction, the even sublattice carries spinor components with $\Gamma(\eta)$ matrices of even rank (including I_4), which give the left-handed components under Lorentz decomposition. The odd sublattice carries the right-handed components. The partition structure and the spinor decomposition are the same object. □

Remark (the checkerboard is not a free choice). The checkerboard partition is not selected by hand to produce chirality — it is the bipartite structure of the lattice itself. The wave equation is nearest-neighbor, so the cubic lattice is bipartite: even sites couple only to odd sites and vice versa. This bipartite structure is a property of the dynamics (range-1 coupling on a hypercubic graph), not of the partition. The staggered-to-Dirac reconstruction identifies the two sublattices with left- and right-handed spinor components — a standard result in lattice QFT [5, 12], not an OI construction. Theorem 12 observes that the OI partition (visible vs. hidden) and the staggered partition (left vs. right) are the same decomposition. The physical partition (the cosmological horizon) traces out a spatial region containing both sublattices; at scales large compared to ϵ, any macroscopic region contains equal numbers of even and odd sites. Chirality in the emergent description is a property of how the staggered-to-Dirac reconstruction organizes the lattice variables into spinor components — it follows from the bipartite structure of the dynamics, not from a choice of which sites to trace out.

7.8 Chirality from the trace-out

Theorem 13 (Chirality). *The emergent gauge coupling of the visible sector is chiral: SU(2) is chiral while SU(3) remains vector-like.*

Proof. (i) *Chirality = sublattice parity.* The staggered-to-Dirac reconstruction [12, 5] assigns spinor components to hypercube corners $\eta \in \{0,1\}^{d+1}$ via $\chi(y+\eta\epsilon) = \frac{1}{4}\Gamma(\eta)_{\alpha\beta}\psi_\beta(y)$, where $\Gamma(\eta) = \gamma_0^{\eta_0}\cdots\gamma_d^{\eta_d}$. Under γ_5: $\Gamma(\eta) \to (-1)^{|\eta|}\Gamma(\eta)$. Sites with $|\eta|$ even are left-handed; $|\eta|$ odd are right-handed. This equals the sublattice parity $\varepsilon(\mathbf{x}) = (-1)^{\sum x_\mu}$. (ii) $D_{LL} = D_{RR} = 0$. The staggered Dirac operator couples only across sublattices: $\{D_{\text{st}}, \varepsilon\} = 0$ (chiral symmetry, Theorem 3), so the even-even and odd-odd blocks vanish identically. (iii) *Trace-out removes right-handed fields.* The OI checkerboard partition assigns even sites (left-handed) to V and odd sites (right-handed) to H (Theorem 12). The trace-out marginalizes over H: all operators in the emergent description act on the visible (left-handed) Hilbert space. (iv) *Left-handed*

effective Lagrangian. Since $D_{LL} = 0$, the gauge coupling in the full theory is purely $L \leftrightarrow R$. After integrating out R (the hidden sublattice) via the Schur complement — a standard procedure in lattice QFT [5, 12] — the effective coupling has the form $\mathcal{L}_{\text{eff}} = \bar{\psi}_L (D_{LR} G_{RR} D_{RL}) \psi_L + \cdots$, where G_{RR} is the hidden-sector propagator. Both external legs are left-handed. (v) *SU(3) vector-like; SU(2) chiral.* The chirality projection P_L acts on the spin/sublattice index and commutes with the internal (K-component) index. SU(3) acts on internal $T_1(3)$ components: both L and R carry color, so the trace-out preserves color coupling → **vector-like**. SU(2) acts on internal $E(2)$ components, but the effective coupling has only L external legs → **chiral**. □

7.9 Anomaly cancellation and hypercharges

Theorem 14 (U(1)_Y automaticity). *The existence of a U(1) gauge factor is automatic in the multi-component framework.*

Proof. Each factor of the commutant decomposes as $\mathrm{U}(n_i) = \mathrm{SU}(n_i) \times \mathrm{U}(1)_i$. For K = 6 with multiplicities (3, 2, 1): $G = [\mathrm{SU}(3) \times \mathrm{U}(1)_B] \times [\mathrm{SU}(2) \times \mathrm{U}(1)_L] \times \mathrm{U}(1)_S$. Three U(1) factors exist automatically. A general abelian charge is $Q = \alpha B + \beta L + \gamma S$. Given SU(3) × SU(2) and the minimal fermion content (5 representations per generation), the six anomaly conditions impose 4 independent constraints on the 5 hypercharge unknowns, leaving exactly one free parameter (overall normalization). There is therefore exactly one anomaly-free U(1). □

Theorem 15 (Unique hypercharges). *Given* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ *with fermions in fundamental or singlet representations, the six anomaly conditions [11] uniquely determine the hypercharges* [10]:

$$Y_Q = \tfrac{1}{6}, \quad Y_u = \tfrac{2}{3}, \quad Y_d = -\tfrac{1}{3}, \quad Y_L = -\tfrac{1}{2}, \quad Y_e = -1$$

Corollary. |q_p| = |q_e| is a theorem, not a coincidence.

7.10 Renormalizability

Theorem 16. *The Yang-Mills coupling has mass dimension [g] = (3−d)/2; dimensionless iff d = 3.*

Proof. The Yang-Mills action $S = \frac{1}{4g^2} \int d^{d+1}x \, F_{\mu\nu}^2$ requires $[g^{-2}][F^2] = [x]^{-(d+1)}$. With $[F] = [x]^{-2}$: $[g]^{-2} = [x]^{d-3}$, so $[g] = (3-d)/2$. □

Since d = 3 is derived (§6), the emergent QFT supports renormalizable gauge interactions with asymptotic freedom.

7.11 Spontaneous symmetry breaking and the hierarchy

Chiral symmetry (Theorem 3) combined with the center-independent gauge sector forbids explicit masses. The unique renormalizable unitarity-preserving mechanism is the Higgs [6, 7]. The minimal scalar breaking $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \to \mathrm{U}(1)_{\text{em}}$ while preserving $\mathrm{SU}(3)_c$ is $H = (\mathbf{1}, \mathbf{2}, +\frac{1}{2})$.

The hierarchy problem is dissolved: in the OI framework, the QFT is emergent. The Higgs mass is set by eigenvalues of M (properties of φ), not by radiative corrections.

7.12 The filter chain

The complete derivation from (S, φ) to the Standard Model is tabulated in Appendix B. The primary chain is proved end-to-end: wave equation → KG → staggered fermions → chiral symmetry → K = 2d = 6 → cubic decomposition (3,2,1) → local gauge invariance → anomaly cancellation → chirality → three generations → one Higgs → θ = 0. Redundant second-route confirmations (asymptotic freedom, minimal N_c, minimal chiral group) are at proposition level and provide independent checks.

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8. The Strong CP Problem

8.1 T-invariance of the wave equation

Theorem 17. *The discrete wave equation is invariant under time reversal T: φ(n, t) → φ(n, −t).*

Proof. Under T, define ψ(n, t) = φ(n, −t). Substitution gives ψ(n, t+1) = ψ(n−1, t) + ψ(n+1, t) − ψ(n, t−1): the same wave equation. □

The checkerboard partition marginalizes over spatial sites, not temporal data. It preserves T while breaking P (§7.8).

8.2 θ = 0

Theorem 18. *The QCD θ-parameter vanishes: θ = 0.*

Proof. The θ-term $\mathcal{L}_\theta = (\theta/32\pi^2)\mathrm{Tr}(F\tilde{F})$ is T-odd: $\mathrm{Tr}(F\tilde{F}) = 2\,\mathbf{E}\cdot\mathbf{B}$, and under T, $\mathbf{E} \to \mathbf{E}$, $\mathbf{B} \to -\mathbf{B}$. A T-invariant Lagrangian cannot contain a T-odd term. The emergent Lagrangian inherits T-invariance because the partition is spatial, so time reversal commutes with the visible/hidden projection. □

8.3 Detailed balance and θ = 0

Theorem 19 (Detailed balance). *For a spatial partition (one preserved by time reversal: $\pi_V \circ \mathcal{T} = \pi_V$), the visible-sector transition probabilities satisfy $T_{ij} = T_{ji}$ for all i, j.*

Proof. The full dynamics φ is a bijection, and T-invariance (Theorem 17) gives $\varphi^{-1} = \mathcal{T} \circ \varphi \circ \mathcal{T}$. For each hidden state h with $\pi_V(\varphi(i,h)) = j$, let $h' = \pi_H(\varphi(i,h))$. Then $\varphi(i,h) = (j,h')$, so $\varphi^{-1}(j,h') = (i,h)$. Since the partition is spatial, $\mathcal{T}$ preserves the visible/hidden classification. The map $h \mapsto h'$ is an injection from $\{h : \pi_V(\varphi(i,h)) = j\}$ into $\{h' : \pi_V(\varphi(j,h')) = i\}$. Since φ is a bijection on a finite set, the injection is a bijection between sets of equal cardinality. Dividing by $|\mathcal{C}_H|$: $T_{ij} = T_{ji}$. □

Theorem 20 (Detailed balance implies T-invariant Hamiltonian). *If $T_{ij} = T_{ji}$ for all i, j, then the emergent Hamiltonian satisfies $[\hat{H}_{\text{eff}}, \hat{\Theta}] = 0$ where $\hat{\Theta}$ is the anti-unitary time-reversal operator.*

Proof. The Born rule gives $T_{ij}(t) = |U_{ij}(t)|^2$ [1, §3.1]. Combined with $T_{ij} = T_{ji}$: $|U_{ij}|^2 = |U_{ji}|^2$ for all i, j, t. This symmetry of the transition amplitudes, by Wigner's theorem, implies $U(t) = \hat{\Theta}\, U(t)^* \,\hat{\Theta}^{-1}$ for a suitable anti-unitary $\hat{\Theta}$, which gives $[\hat{H}_{\text{eff}}, \hat{\Theta}] = 0$. □

Theorem 21 ($\bar{\theta} = 0$ at all energy scales). *The physical parameter $\bar{\theta} = \theta + \arg\det(Y_u Y_d) = 0$, and the vanishing persists to all infrared scales.*

Proof. **Step 1:** Since φ is a bijection, φ^n is a bijection for every n. Applying the counting argument of Theorem 19 to φ^n: $T_{ij}(n) = T_{ji}(n)$ for all n. (Verified numerically: T-violation is exactly zero for L = 4, 6, q = 2, 3, 5, and all n = 1, ..., 12.)

Step 2: By Theorem 20 applied to $T_{ij}(n)$ at each scale: $[\hat{H}_{\text{eff}}(n), \hat{\Theta}] = 0$. The effective Hamiltonian is T-invariant at every time scale.

Step 3: T-invariant Yukawa couplings at each scale are simultaneously real in an appropriate basis [10, §6.3]. Real Yukawa matrices have $\arg\det(Y_u Y_d) = 0$. Combined with θ = 0 (Theorem 18): $\bar{\theta} = 0$ at every scale. □

Remark. This proof bypasses the instanton question. T-invariance of the transition probabilities is an exact consequence of the bijection structure, holding at every time scale without perturbative or non-perturbative approximation. The RG flow cannot generate T-violation because the underlying bijection structure forbids it at every scale.

Prediction: $\bar{\theta} = 0$ exactly. No axion needed. Neutron EDM should be exactly zero (current bound: $|d_n| < 1.8 \times 10^{-26}\ e \cdot$ cm [22]).

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9. Discussion

9.1 Structural realism

The structural reading aligns with ontic structural realism but does not require it. What the framework proves is that (S, φ) is a *complete description* of reality up to gauge equivalence (the reconstruction theorem, §10.3). Whether the structure *is* reality (the OSR commitment) or *describes* a reality that exists independently is a question the framework identifies as gauge — provably undecidable by any measurement. The "stuff" of the universe, in any reading, is fully characterized by the abstract structure of a bijection on a finite set with bounded coupling.

9.2 The ontological hierarchy

The triple (S, φ, V) generates every concept in fundamental physics, not as independent substances but as different aspects of the same structure. **Space** is the coupling

structure of φ — the graph G_φ determined by which degrees of freedom affect which others (§3.1). **Matter** is the state — localized patterns that propagate through the coupling graph. **Energy** is the rate of change under iteration. **Time** is the iteration itself. **Quantum mechanics** is the observer's compressed description of the visible sector. **General relativity** is the thermodynamic limit of the coupling structure. **Conservation laws** are emergent: energy conservation (Noether) is what information conservation (bijectivity) looks like in the emergent quantum description. None of these are independent entities; they are descriptions of (S, φ, V) at different scales.

9.3 The measurement problem

On the structural reading, the measurement problem is dissolved. The wave function is not a component of (S, φ, V) — it is a derived object. Since it is derived, not fundamental, asking "does it collapse?" is asking about the behavior of a compression artifact. In the double-slit experiment, the particle traverses a single slit in the deterministic substratum. In Wigner's friend, the Friend has a definite outcome; Wigner's superposition reflects his epistemic deficit.

Branching is forbidden by the rigidity of φ. A fixed bijection on a finite set has exactly one trajectory from any initial state. There is no point at which the trajectory splits. The appearance of branching in the emergent quantum description reflects the observer's uncertainty about which trajectory they are on (because they cannot see the hidden sector), not a physical splitting of worlds.

Non-locality in Bell correlations is explained by the coupling graph G_φ. Two visible sites prepared in a joint state (entangled) have correlated hidden-sector configurations — a consequence of the joint P-indivisible dynamics at preparation [1, §3.2]. The correlations are mediated by the coupling graph, not by any superluminal influence. The graph structure ensures that the correlations respect the Tsirelson bound and violate Bell inequalities without violating parameter independence.

9.4 Observer genericity

Theorem 22 (Genericity of observers). *Let φ be the wave equation (or any energy-conserving dynamics) on a connected bounded-degree coupling graph with diameter D ≥ 4. Then for any connected subgraph V with $|V| \leq N/3$, the partition (V, H) satisfies C1–C3.*

Proof. C1: G_φ is connected and V is a proper subgraph, so ∂V ≠ ∅. The wave equation couples nearest neighbors; any edge in ∂V produces non-trivial dynamical coupling.

C2: the wave equation is a Hamiltonian system — it conserves energy and preserves phase-space volume. The hidden sector has spectral gap Δ = 0: correlations persist indefinitely, not just until some relaxation time. The C2 necessity proof [1, §3.3] establishes that $\Delta\tau_S \gg 1$ drives the system to Markovianity; for any Hamiltonian hidden sector, $\Delta\tau_S = 0 \ll 1$ — maximally in the slow-bath regime ($\tau_B \to \infty$, the strongest possible satisfaction of $\tau_S \ll \tau_B$), regardless of partition geometry.

C3: $|H| = N - |V| \geq 2N/3$. The hidden-sector configuration space has $q^{|H|}$ states; the non-Markovian mutual information is bounded by $|H| \log_2 q$ bits [1, §3.3]. The

capacity ratio $q^{|H|-|V|} \geq q^{N/3}$ is exponentially large. No visible-sector process of sub-exponential duration saturates the hidden sector's memory. □

Remark (validity window). While C2 is satisfied maximally, the emergent Hamiltonian has a finite *stationarity window*: the time before information deposited at the V–H boundary propagates through H and returns to V at a different boundary point. The wave equation has an exact light cone (no exponential tails), so the minimum return time is $\tau_{\text{return}} = \min_{b,b' \in \partial V} \text{dist}_H(b, b')$. For core-shell partitions with core radius r on a lattice of side L: $\tau_{\text{return}} = (L - 2r)/v$. After τ_{return}, the emergent description remains quantum mechanical (C2 is still satisfied), but the emergent Hamiltonian becomes time-dependent. Numerical verification: for L = 20–320, τ_{return} ranges from 6 to 318 steps, confirming the light-cone bound.

Corollary. Any finite deterministic system with energy-conserving dynamics and bounded-degree coupling necessarily contains subsystems whose internal description is quantum mechanics. The observer is not postulated — it is a mathematical consequence.

9.5 Relation to other approaches

Causal set theory. The bijection φ with coupling graph G_φ generates a causal partial order that is a causal set in the sense of Bombelli-Lee-Meyer-Sorkin [4]. The OI framework provides what CST lacks: a deterministic dynamics that provably produces both QM and GR. The CST Hauptvermutung corresponds to the Myrheim-Meyer dimension result [4].

't Hooft [17]. Agrees on emergent QM from determinism but differs: OI requires bijectivity (not information loss), violates outcome independence (not superdeterminism), and derives $\hbar$, S_BH, and Einstein's equations.

Wolfram [18]. Shares discrete dynamics and emergent space but differs: OI requires bounded coupling for the area law and produces quantitative predictions for physical constants.

Loop quantum gravity [19]. Both find discrete spatial geometry at the Planck scale, but LQG quantizes GR (starts from the continuum and discretizes), while OI derives both QM and GR from a pre-quantum starting point.

9.6 Predictions

The framework makes the following falsifiable predictions.

From the cosmological application [1]: dark energy evolution with $\nu_{\text{OI}} \approx 2.45 \times 10^{-3}$ (DESI, Euclid); GW echoes near black hole horizons (LIGO/Virgo/KAGRA).

From the SM derivation (§§7–8): no additional gauge groups below M_{Pl}; no fundamental scalars beyond the Higgs doublet; no SUSY partners; no fourth generation; $\bar{\theta} = 0$ exactly (no axion, neutron EDM = 0); the CKM CP-violating phase arises from P-violation (the partition), not from T-violation; neutrinos are Majorana; no proton decay from GUT-scale gauge bosons (the eigenvalues of M correspond to independent irreps of O, so couplings do not unify).

The conjunction is distinctive to this framework.

9.7 What the framework determines and what the specific bijection determines

The framework's predictions fall into two categories, and the distinction matters for assessing their status.

Structural predictions are properties of the equivalence class [(S, φ)]/$\mathcal{G}_{\text{sub}}$ — they hold for every bijection satisfying the six structural properties. These include: SU(3) × SU(2) × U(1) with multiplicities (3, 2, 1), three chiral generations with identical gauge quantum numbers, one Higgs doublet (1, 2, +1/2), unique hypercharges from anomaly cancellation, $\bar{\theta} = 0$ at all energy scales, $\hbar = c^3\varepsilon^2/(4G)$, and the Bekenstein-Hawking formula. These are theorems. They do not depend on which specific φ describes our universe.

Solution-specific properties are determined by the particular bijection φ within the equivalence class. These include: gauge couplings (set by the eigenvalues μ_c, μ_w of M), fermion masses (set by taste-breaking in φ at $\mathcal{O}(\epsilon^2)$), and mixing parameters (set by the Yukawa structure of φ). These are analogous to the mass M in the Schwarzschild solution: GR derives the functional form of the metric but does not predict which M describes a particular black hole. The OI framework derives the SM's structure but does not predict which φ describes our universe.

The "mass hierarchy" — the five-order-of-magnitude spread of fermion masses — falls in the second category. It is not a gap in the framework; it is a property of the specific solution. The framework makes the correct structural prediction: three generations with *identical* gauge quantum numbers (from cubic symmetry) but *non-degenerate* masses (because the specific φ breaks the cubic symmetry through its Yukawa structure). The hierarchy problem proper — why the Higgs mass is so much lighter than the Planck scale — is dissolved: the Higgs mass is an eigenvalue of M (a property of φ), not a radiative correction within the emergent QFT (§7.11).

Remaining open problems. (i) *Lattice-continuum comparison* — the structural predictions are exact theorems at the lattice level and do not require a continuum limit. The lattice is the fundamental description ($\epsilon = 2l_p$ is physical, not a regulator), so there is no $\epsilon \to 0$ limit to take. The comparison to experiment — which is interpreted through continuum perturbation theory — introduces corrections of order $(E/M_{\text{Pl}})^2 \sim 10^{-32}$ at LHC energies, far below any foreseeable experimental sensitivity. The Yang-Mills mass gap problem (proving rigorous existence of the continuum limit for Yang-Mills theory) is a famous open problem in mathematics, but it does not affect any prediction of the OI framework: the lattice theory is complete as stated, and the lattice is the fundamental description, not a regularization. (ii) *Generation identification* — the fermion count (three) is proved; the assignment of specific representations to specific tastes is a uniqueness argument from anomaly cancellation, not a geometric derivation.

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10. Synthesis: The Complete Program

10.1 Summary of results

The framework starts from four axioms (deterministic dynamics, finiteness, causal partition, classical probability) and three conditions (C1–C3), with no quantum mechanics, general relativity, or specific dynamics. The companion paper [1] proves that embedded observation under these conditions is equivalent to unitary QM, derives $\hbar = c^3\varepsilon^2/(4G)$ and the Bekenstein-Hawking formula, dissolves the CC discrepancy, and predicts dark energy evolution and GW echoes. This paper establishes that center independence, isotropy, and linearity uniquely select the wave equation (§7.1), which passes all seven GR tests with no free parameters (§5). The framework's results depend on six structural properties and nothing else; the wave equation on a d = 3 lattice determines the Standard Model with the primary derivation chain proved end-to-end (§7).

Theorem dependency structure. The 24 theorems are not a linear chain of 24 sequential steps. They form a tree with four independent branches. The critical path through the SM derivation is five theorems long (1 → 2 → 5 → 6 → 7); the strong CP solution is independent of the SM structure; and the reconstruction uses all preceding results but adds only two new theorems.

```
Axioms + d=3
  |
  ├─→ Thm 1 (wave eq) ─→ Thm 2 (Dirac) ─→ Thm 3 (chiral sym)
  |         |
  |         ├─→ Thm 4 (mass spectrum)
  |         └─→ Thm 5 (gauge = commutant of M)
  |
  ├─→ Thm 6 (K=2d) ─┐
  |                 ├─→ Thm 7 (cubic decomposition → multiplicities (3,2,1))
  |                 |     |
  |                 |     ├─→ Thms 8–11 (tastes, generations, Higgs)
  |                 |     ├─→ Thm 13 (chirality)
  |                 |     └─→ Thms 14–15 (hypercharges, anomaly cancellation)
  |                 |
  |                 └─→ Thm 16 (renormalizability iff d=3)
  |
  ├─→ Thm 12 (partition = spinor) [independent of Thms 6–11]
  |
  ├─→ Thms 17–21 (strong CP: θ̄ = 0) [independent of Thms 6–16; uses only bijectivity]
  |
  ├─→ Thm 22 (observer genericity) [independent of Thms 1–21]
  |
  └─→ Thm 23 (reconstruction) [uses Thms 1–22]
        └─→ Thm 24 (substratum gauge group) [uses Thm 23]
```

Each branch can be verified independently. The strong CP block (Thms 17–21) requires only the bijectivity of φ and the wave equation's T-invariance — it does not depend on the gauge structure derivation. The observer genericity theorem (Thm

22) is independent of all preceding results.

10.2 The physical interpretation of (S, φ)

Storage and memory. S is the set of all distinguishable states: *finite capacity*. φ is a bijection: *perfect memory* — information is never created or destroyed. Together, (S, φ) is a finite lossless memory. The partition V defines the observer's *access* — both read and write. The observer reads the visible sector (measuring x) and writes to the hidden sector (each visible-sector operation imprints correlations on H through the coupling H_{int}). The slow bath (C2) preserves those writes; subsequent reads retrieve them — producing the information backflow that constitutes P-indivisibility. Quantum mechanics is the statistics of this read-write cycle: not passive observation of a static memory, but the self-consistent description that emerges when a finite subsystem both reads from and writes to a lossless register it cannot fully access. The Born rule is the equilibrium of the cycle, not of passive reading. Interference is write-then-read: information deposited in the hidden sector during one transition is retrieved on a later transition and partially cancels or reinforces the transition probabilities.

The 10^{122} CC discrepancy is the compression ratio between total storage and readable storage. The dark sector is the gravitational effect of the unreadable storage. The Bekenstein-Hawking entropy is the storage capacity of the partition boundary. The action scale $\hbar$ is the conversion factor between storage geometry and read-write statistics.

The substrate objection. "What is the memory made of?" (S, φ) is a *complete description* of reality — it determines all observables. Space, time, matter, and energy are derived from (S, φ), so they cannot appear in its definition without circularity. Whether (S, φ) *is* reality or *describes* reality is provably undecidable by any measurement (reconstruction theorem below).

Relation to computation. A Turing machine has a tape (storage), a head (partial read/write access), and a transition function (update rule). The correspondence is suggestive: S is the tape, V is the head's access window, φ is the transition function. But three differences are physically significant. First, a Turing machine's tape is potentially infinite; S is finite — finiteness is essential for the recurrence proof of P-indivisibility and QM, though the effective finiteness result (Main, §9.6) shows the deep hidden sector may be infinite without affecting the observable physics. Second, a Turing machine is generally irreversible (it can erase, overwrite, and halt); φ is a bijection — nothing is erased, nothing is created, there is no halting. Third, a Turing machine computes an *extrinsic function* (input → output for an external user); (S, φ) computes no extrinsic function — it is a closed permutation that cycles through states and returns. The appearance of dynamics, probability, particles, and forces is entirely the observer's perspective — what the permutation looks like through the partial window V.

Remark (the Turing connection under effective finiteness). The three differences soften under the effective finiteness result. With the deep hidden sector potentially infinite, the first difference is between the formal axiom (S finite) and the physical requirement (only $\mathcal{C}_V \times \mathcal{C}_B$ need be finite). The second difference is a specialization, not an opposition: reversible Turing machines (Bennett, Fredkin-Toffoli)

are a well-studied subclass. The third difference — extrinsic vs. intrinsic — is the one that does physical work. The mapping is then structural: V is the head, H is the tape, φ is a reversible transition function, and C1–C3 characterize the architecture. The framework extends Turing's question: instead of asking what a machine can compute for an external observer, it asks what computation looks like to a component of the machine — and proves the answer is quantum mechanics.

The arrow of time. The substratum has no arrow of time — φ and φ^{-1} are equally valid. Entropy increase is a property of the observer's coarse-grained description: the standard Boltzmann mechanism applied to the partition.

The incompleteness family. The framework's central result belongs to a family of impossibility-with-structure theorems. Gödel: a formal system cannot prove all truths about itself — the unprovable truths have rigid structure. Turing: a computer cannot decide all questions about its own behavior — the undecidable problems have rigid structure. OI: an observer embedded in a deterministic system cannot access the complete state — the emergent description has rigid structure (quantum mechanics). The common structure is *self-reference under finite resources*.

The precise OI analog of the halting problem is: *can the observer determine the hidden-sector state h?* The question is well-posed — h has a definite value at every moment, because (S, φ) is deterministic. Different h values produce different physical outcomes. But the observer provably cannot determine h: multiple hidden states are compatible with any visible-sector history, and transition probabilities $T_{ij}(t)$ are averages over h. The structural consequence of this inaccessibility is quantum mechanics — just as the structural consequence of the halting problem's undecidability is computability theory. The framework also identifies a second class of inaccessible quantities — the alphabet size q (§4) and the deep-sector cardinality $|\mathcal{C}_D|$ (Main, §5.2) — but these are *gauge*, not undecidable: different values produce identical observables, so the question itself is physically empty. The hidden state h is undecidable (real answer, provably inaccessible). The cardinality $|\mathcal{C}_D|$ is gauge (no answer to find).

Mathematics and physics. The trace-out performs a Jordan-Chevalley projection (Appendix A): it extracts the semisimple part of the dynamics and erases the nilpotent monodromy. Physics is the semisimple shadow of mathematics — the diagonalizable spectral data, projected by the trace-out and organized by the gauge group's representation structure. The reconstruction theorem (below) proves that the mathematical description and the physics determine each other uniquely up to gauge.

10.3 The reconstruction theorem

The forward direction — from (S, φ) to observed physics — is established by §§5, 7–8 and the companion paper [1]. The converse question is whether the observed physics uniquely determines (S, φ). The reconstruction proceeds in three stages.

Stage 1: From observed QM to deterministic embedding.

Empirical input: Unitary quantum mechanics with Bell violations, on a configuration space with finite entropy bounded by a finite-area surface.

The finite-entropy condition has independent justification from holographic bounds [3, 20]. The cosmological horizon has finite area; the bound applies.

Derivation. Any CPTP quantum channel on a finite-dimensional Hilbert space admits a Stinespring dilation as a bijection on a finite set [1, Appendix A]. Bell violations require non-trivial coupling (C1). The characterization theorem [1, §3.3] requires slow-bath memory (C2) and sufficient capacity (C3).

Output: The observed QM arises from some (S, φ, V) satisfying C1–C3. *Status:* Theorem.

Stage 2: From the embedding to the dynamics and gauge structure.

Additional empirical input: Spatial isotropy.

Derivation. The coupling graph gives d = 3 (§6). Center independence + isotropy + linearity uniquely select the wave equation (§7.1). The wave equation determines the full SM structure: K = 2d = 6 (Theorem 6), cubic decomposition (3,2,1) (Theorem 7), local gauge invariance (§5), staggered species → three generations + one Higgs (Theorems 8–11), partition-spinor identification (Theorem 12), chirality (Theorem 13), anomaly cancellation (Theorems 14–15), T-invariance → θ = 0 (Theorems 17–21).

Output: Dynamics, dimension, gauge group, matter content, and discrete symmetries are all derived. The SM structure is a retrodiction checked against observation. *Status:* Theorem at the lattice level.

Stage 3: From the dynamics to the emergent constants.

No additional input. The gap equation from thermal self-consistency [1, §5] gives $\hbar = c^3\epsilon^2/(4G)$ with ε = 2 l_p. *Status:* Theorem.

Theorem 23 (Layered reconstruction). *Observed quantum mechanics with Bell violations and finite boundary entropy, together with spatial isotropy, uniquely determine — at the lattice level — the equivalence class $[(S, \varphi)]/\sim$: a finite set with a bijection of bounded coupling degree and statistical isotropy, with d = 3, K = 6, coupling matrix eigenvalue multiplicities (3, 2, 1), $\bar{\theta} = 0$, and $\hbar = c^3\epsilon^2/(4G)$.*

Proof. Stage 1 ⇒ Stage 2 ⇒ Stage 3, as detailed above. □

The theorem starts from empirical inputs (QM, Bell violations, finite entropy, isotropy) and derives the framework. The SM gauge structure, three generations, one Higgs, θ = 0, and the Bekenstein-Hawking formula are retrodictions — derived, not assumed.

Remark (Evidential weight of the reconstruction). The reconstruction uniquely derives $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ with multiplicities $(3, 2, 1)$, three chiral generations, one Higgs doublet $(1, 2, +1/2)$, unique hypercharges, and $\bar{\theta} = 0$. Because the derivation is unique — no free parameters, no model-building choices, no landscape of alternatives — every independent experimental confirmation of the Standard Model's gauge structure is retroactive evidence for the derivation chain that produces it. The Standard Model is the most precisely tested mathematical structure in the history of science ($g - 2$ of the electron confirmed to 10^{-12}, electroweak precision observables to 10^{-5}, QCD cross-sections across decades of energy). This experimental record

transfers in full to the framework, because uniqueness of the derivation makes the relationship between evidence and theory transitive: data confirms structure, structure is uniquely derived, therefore data confirms the derivation.

Remark (Scope). The reconstruction establishes uniqueness within the class of finite deterministic systems with bounded coupling. It does not exclude intrinsically continuum or stochastic alternatives.

Remark (Continuum extension). The lattice-level predictions are the framework's primary claims — the lattice is the fundamental description, not an approximation to a continuum theory. The structural results (gauge group, representations, generation count, $\bar{\theta} = 0$) are algebraic/topological properties of the lattice theory and are exact. Quantitative observables (scattering amplitudes, mass ratios) are compared to experiment through continuum perturbation theory; the lattice-continuum discrepancy at any experimentally accessible energy E is suppressed by $(E\epsilon/\hbar c)^2 = (E/M_{\text{Pl}})^2$, which is $\sim 10^{-32}$ at LHC energies. The rigorous proof that lattice Yang-Mills defines a continuum limit with a mass gap (a Clay Millennium Prize problem) is an open problem in mathematics, but it is not required by the OI framework: the lattice theory is complete, and its predictions are lattice-exact.

The reconstruction establishes a bidirectional correspondence:

$$\text{Observed physics (QM + Bell + finite entropy + isotropy)} \quad \longleftrightarrow \quad [(S, \varphi)]/\sim$$

The mathematical structure and the physics determine each other up to gauge equivalence. The distinction between "mathematics describes reality" and "mathematics is reality" has no empirical content — it is itself gauge. This reframes Wigner's puzzle: the "unreasonable effectiveness" of mathematics is a theorem, not a mystery.

10.4 The substratum gauge group

The equivalence relation $\sim$ in the reconstruction theorem has a precise structure. Define: $(S, \varphi) \sim (S', \varphi')$ if the two systems produce identical emergent physics — the same transition probabilities $T_{ij}(t)$, the same emergent Hamiltonian (up to D-gauge), and the same $\hbar$ — for all partitions of the same structural class. The set of transformations mapping (S, φ) to an equivalent (S', φ') is a group — the *substratum gauge group* $\mathcal{G}_{\text{sub}}$.

Theorem 24 (Generators of the substratum gauge group). *$\mathcal{G}_{sub}$ contains at least four independent families of transformations:*

(i) State relabeling. For any bijection $\sigma : S \to S$, the conjugate $(S, \sigma \circ \varphi \circ \sigma^{-1}) \sim (S, \varphi)$. The transition probabilities depend on the coupling structure of φ, not on which labels are attached to which states. This is an $|S|!$-element subgroup — vastly larger than any gauge group in the Standard Model.

(ii) Alphabet change. Replacing the local state space $\mathbb{Z}/q\mathbb{Z}$ with $\mathbb{Z}/q'\mathbb{Z}$ for any $q' \geq 2$, while preserving the coupling graph and dynamics class, leaves all observables unchanged (§4). This family is parametrized by all integers $q \geq 2$.

(iii) Deep-sector enlargement. Adjoining additional degrees of freedom to $\mathcal{C}_D$ (the deep hidden sector beyond the boundary layer), with arbitrary dynamics satisfying $\tau_B^D \gg \tau_S$, does not change the emergent description. The boundary-only dependence lemma [1, §5.2] proves $T_{ij}(t) = T_{ij}^{(B)}(t) + \mathcal{O}(t/\tau_B)$: observables depend only on $\mathcal{C}_V \times \mathcal{C}_B$. The deep sector may be finite of any size, or infinite.

(iv) Graph isomorphism (up to statistical isotropy). Two coupling graphs G_φ and $G_{\varphi'}$ that are quasi-isometric with the same polynomial growth exponent d, the same spectral properties, and the same statistical isotropy at large scales produce the same emergent physics. The regular cubic lattice $\mathbb{Z}^3$ and any bounded-degree random graph with $d = 3$ polynomial growth and statistical isotropy are gauge-equivalent.

Proof. Each generator preserves all inputs to the derivation chain. (i): conjugation preserves the coupling graph G_φ up to relabeling, hence all graph-dependent quantities (area law, dispersion, dimension, eigenvalue multiplicities). (ii): §4 proves q-independence of every prediction. (iii): the boundary-only dependence lemma gives the result directly. (iv): the derivation chain uses only statistical properties of G_φ (dimension via Myrheim-Meyer, isotropy, bounded degree), not the specific graph. □

Completeness. The reconstruction theorem (Theorem 23) establishes that the map $\Phi : (S, \varphi) \mapsto$ (observables) has fibers $[(S, \varphi)]/\sim$. By definition, $\mathcal{G}_{\text{sub}}$ is the stabilizer of Φ — the set of transformations that preserve all observables. A transformation $g \notin \mathcal{G}_{\text{sub}}$ would, by definition, change at least one observable, placing $g \cdot (S, \varphi)$ in a different fiber. Since Theorem 23 proves the fibers are uniquely determined, the four generators — which exhaust all transformations preserving the six structural inputs to the derivation chain (coupling structure, dimensionality, spectral properties, boundary geometry, entropy counting, dynamics class) — span the full gauge group. $\mathcal{G}_{\text{sub}}$ is the kernel of the reconstruction map.

The gauge hierarchy. Three levels of gauge symmetry appear in the framework, each projecting onto the next through the trace-out:

Level 3 (substratum): $\mathcal{G}_{\text{sub}}$ acts on (S, φ) before the trace-out. It is the largest gauge group and includes transformations with no analog in the emergent description (deep-sector enlargement, alphabet change).

Level 2 (emergent QFT): $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ is the commutant of the coupling matrix M, acting on the emergent fields. It is the image of $\mathcal{G}_{\text{sub}}$ restricted to transformations that permute internal components within the eigenspaces of M.

Level 1 (emergent Hamiltonian): The D-gauge $H \to DHD^\dagger$ with D a diagonal unitary, acting on the emergent Hamiltonian within the emergent QM. It is the residual freedom after all transition-probability data has been extracted.

Each level is contained in the one above: Level 1 $\subset$ Level 2 $\subset$ Level 3. The trace-out projects Level 3 onto Level 2 (the SM gauge group is the shadow of $\mathcal{G}_{\text{sub}}$ visible to the emergent QFT), and restricting to the Hamiltonian projects Level 2 onto Level 1.

Remark. The substratum gauge group is not a symmetry of a Lagrangian or an action — no Lagrangian exists at the substratum level. It is a symmetry of the *equivalence class of substrata*, defined by the condition that all observables are preserved. The

emergent gauge symmetries (Levels 1 and 2) are Lagrangian symmetries in the standard sense, derived from the substratum through the trace-out.

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11. Conclusion

The lattice in the OI framework is not a physical substrate. It is the coupling graph of a bijection on a finite set. Space is the large-scale geometry of this graph. Time is the iteration of the bijection. Quantum mechanics is the observer's compressed description of the visible sector. General relativity is the thermodynamic limit of the coupling structure. The Standard Model is the specific QFT determined by the lattice structure in d = 3.

The fundamental object is (S, φ) — a finite set and a bijection. Physically: a finite lossless memory. The partition V is derived (any small subgraph sees QM). The dimension d = 3 is derived (four independent filters converge). The gauge group, matter content, and discrete symmetries are derived (the primary chain is proved end-to-end). The reconstruction theorem (§10.3) establishes the converse: observed physics uniquely determines $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$ at the lattice level. The universe is completely described by a lossless memory with finite read-write access. Physics is what that memory looks like from inside — and what it looks like from inside uniquely determines what it is.

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Appendix A: The Trace-Out as a Jordan-Chevalley Projection

This appendix proves that the OI trace-out extracts the semisimple part of the evolution matrix's Jordan-Chevalley decomposition, erasing the nilpotent monodromy. Results are stated for the scalar wave equation on a ring of L sites over $\mathbb{F}_q$; the multi-component extension incorporates the gauge group.

Setup

The wave equation $x(n,t+1) = x(n-1,t) + x(n+1,t) - x(n,t-1) \pmod q$ has phase space $(\mathbb{Z}/q\mathbb{Z})^{2L}$ and evolution matrix $F = \left(\begin{smallmatrix} 0 & I \\ -I & A \end{smallmatrix}\right)$ where A is the circulant adjacency matrix.

Theorem A.1 (Period formula)

For q prime with $\gcd(L,q) = 1$*:* $\mathrm{ord}(F \bmod q) = qL$ *if q is odd, and L if q = 2.*

Proof. For each Fourier mode k, the 2×2 block B_k has characteristic polynomial $t^2 - \lambda_k t + 1$. At parabolic modes ($\lambda_k = \pm 2$), the Jordan form gives $B_k^n = \alpha^n \left(\begin{smallmatrix} 1 & n\alpha^{-1} \\ 0 & 1 \end{smallmatrix}\right)$, contributing order q (or 2q). The diagonalizable eigenvalues contribute orders dividing L. Therefore $\mathrm{ord}(F) = qL$. For q = 2: the nilpotent part is automatically killed. □

Theorem A.2 (Jordan-Chevalley decomposition)

Define $N = (F^L - I)/L \bmod q$*. Then: (i) N is nilpotent with* $N^2 = 0$ *and rank(N) = 2. (ii)* $F_u = I + N$ *is unipotent with* $F_u^q = I$*. (iii)* $F_{\mathrm{ss}} = F \cdot (I - N)$ *is semisimple with* $F_{\mathrm{ss}}^L = I$*. (iv)* $F = F_{\mathrm{ss}} \cdot F_u$ *and* $[F_{\mathrm{ss}}, N] = 0$*.*

Proof. F^L has the form $I + LN'$ where N' arises from the Jordan blocks: $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)^L = \left(\begin{smallmatrix} 1 & L \\ 0 & 1 \end{smallmatrix}\right)$ contributes a rank-1 nilpotent, and similarly for $\alpha = -1$ (since $(-1)^L = 1$ for even L). So $N = N'$ has $N^2 = 0$ (each block is rank-1 nilpotent on a 2D subspace) and $\mathrm{rank}(N) = 2$ (two parabolic modes). Since $N^2 = 0$: $(I+N)^{-1} = I - N$, giving $F_{\mathrm{ss}} = F(I-N)$. Then $F_{\mathrm{ss}}^L = F^L(I-N)^L = (I+LN)(I-LN+\ldots) = I \bmod q$ since terms involving LN cancel modulo q (using $N^2 = 0$). Commutativity follows from N being supported on the parabolic eigenspaces, which are F-invariant. □

Verified computationally for L = 4, 6, 8, 10, 12 and q = 3, 5, 7, 11, 13.

Theorem A.3 (q-independence of the Weil-Deligne conductor)

The Weil-Deligne conductor $\mathfrak{f}_{\mathrm{WD}} = \mathfrak{f}_{\mathrm{ss}}(L) + \mathrm{rank}(N) = \mathfrak{f}_{\mathrm{ss}}(L) + 2$ *is q-independent when* $\gcd(L,q) = 1$*, where* $\mathfrak{f}_{\mathrm{ss}}(L) = \sum_\alpha (\mathrm{ord}(\alpha) - 1)$ *is computed from the eigenvalue orders of* F_{ss}*, all dividing* L*.*

Proof. F_{ss} has order L; its eigenvalues are L-th roots of unity. For $\gcd(L,q) = 1$, the L-th roots in $\bar{\mathbb{F}}_q$ are isomorphic to those in $\mathbb{C}$, so their orders match. □

L	$\mathfrak{f}_{\text{ss}}(L)$	$\mathfrak{f}_{\text{WD}}$	$\text{NM}^2 = 3L/4$
4	14	16	3.00
6	30	32	4.50
8	70	72	6.00
10	106	108	7.50
12	130	132	9.00

Both $\mathfrak{f}_{\text{ss}}$ and NM^2 are q-independent encodings of the same semisimple eigenvalue data — one via multiplicative orders (integers), one via fourth moments of magnitudes (reals).

Theorem A.4 (Additive decomposition over gauge irreps)

For the K-component wave equation with coupling matrix $M = \text{diag}(\mu_1 I_{n_1}, \dots, \mu_r I_{n_r})$*:*

$$\mathfrak{f}_{\text{WD}}(M) = \sum_{i=1}^{r} n_i \cdot \mathfrak{f}_{\text{WD}}(\mu_i)$$

In particular, for K = 6 with $M = \text{diag}(\mu_c I_3, \mu_w I_2, 1)$*:* $\mathfrak{f}_{\text{WD}} = 3\mathfrak{f}_{\text{WD}}(\mu_c) + 2\mathfrak{f}_{\text{WD}}(\mu_w) + \mathfrak{f}_{\text{WD}}(1)$*, with the same multiplicities* $(3, 2, 1)$ *that determine* $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$*.*

Proof. The multi-component evolution matrix is block-diagonal: the a-th component has its own $2L \times 2L$ block F_{μ_a} with eigenvalues depending only on μ_a. Both F_{ss} and N inherit the block-diagonal structure, so $\mathfrak{f}_{\text{ss}}$ and $\text{rank}(N)$ decompose additively. □

Verified for all $(\mu_c, \mu_w) \in \{1, \dots, q-1\}^2$ *at q = 3, 5, 7 with L = 4. Every case matches exactly.*

The projection

The OI trace-out (marginalization over the hidden sector) produces the emergent quantum description, which depends on the coupling eigenvalues μ_k via the NM formula $\text{NM}^2 = 3\langle\mu^4\rangle$ (a consequence of the stochastic-quantum correspondence applied to the wave equation's Fourier decomposition). These μ_k are properties of F_{ss} — the semisimple part of the dynamics. The nilpotent monodromy N contributes nothing to the emergent description: it affects only the off-diagonal Jordan block entries, which are erased by the coarse-graining over the hidden sector.

The trace-out therefore performs the Jordan-Chevalley projection $(F_{\text{ss}}, N) \mapsto F_{\text{ss}} \mapsto \{\mu_k\} \mapsto \text{NM}^2$: it extracts the semisimple part, encodes it via magnitudes rather than orders, and organizes it by the representation theory of the partition. The nilpotent monodromy — genuine mathematical structure present in the full dynamics — is invisible to the embedded observer. This is the precise sense in which physics is the semisimple shadow of mathematics: the observer sees only the diagonalizable spectral data, projected by the trace-out and organized by the gauge group's representation structure.

Appendix B: The Filter Chain

Step	Constraint	Selects	Status
0	Emergent QFT, d = 3	Lattice QFT, 3+1D	Theorem
1	Wave eq. = massless KG (Thm 1)	Staggered fermions	Theorem
2	Center indep. = chiral sym. (Thm 3)	Massless chiral fermions	Theorem
3	Multi-component → matrix WE (§7.4)	Gauge structure from M	Theorem
4	Gauge group = commutant of M (Thm 5)	$\mathrm{U}(n_1) \times \cdots$	Theorem
4a	K = 2d = 6, cubic decomp. (Thms 6–7)	Multiplicities (3,2,1)	Theorem
5	Background indep. → local gauge (§5)	Local gauge invariance	Theorem
6	Dimensionless g (Thm 16)	Renormalizable gauge theories	Theorem
7	Spin-taste + cubic sym. (Thms 8–11)	1 Higgs + 3 gen with SM reps	Theorem
8	Partition = spinor decomp. (Thm 12)	Checkerboard ↔ chirality	Theorem
9	$D_{LL} = 0$ + trace-out (Thm 13)	SU(2) chiral, SU(3) vector-like	Theorem
10	U(1)_Y automaticity (Thm 14)	Unique abelian factor	Theorem
11	Anomaly cancellation (Thm 15)	SM hypercharges	Theorem
12	T-invariance (Thms 17–18)	θ = 0	Theorem
13	Detailed balance at all scales (Thms 19–21)	θ = 0 at all scales	Theorem

Output: SU(3)_c × SU(2)_L × U(1)_Y with:

$$Q_L = (\mathbf{3}, \mathbf{2}, +\tfrac{1}{6}), \quad u_R = (\mathbf{3}, \mathbf{1}, +\tfrac{2}{3}), \quad d_R = (\mathbf{3}, \mathbf{1}, -\tfrac{1}{3})$$
$$L_L = (\mathbf{1}, \mathbf{2}, -\tfrac{1}{2}), \quad e_R = (\mathbf{1}, \mathbf{1}, -1)$$

Three generations. One Higgs doublet $H = (\mathbf{1}, \mathbf{2}, +\frac{1}{2})$. $\bar{\theta} = 0$. Primary chain proved end-to-end. This is exactly the Standard Model.

Appendix C: Non-Markovianity Under Gauge Structure

The multi-component system's P-indivisibility decomposes additively over eigenmodes:

$$\|G_V(2) - G_V(1)^2\|_F^2 = \sum_{a=1}^{K} \|g^{\mu_a}(2) - g^{\mu_a}(1)^2\|_F^2$$

The normalized non-Markovianity has the exact closed form:

$$\mathrm{NM}_{\mathrm{norm}} = \sqrt{3} \cdot \sqrt{\langle \mu^4 \rangle}$$

where $\langle \mu^4 \rangle = \frac{1}{K} \sum_{a=1}^{K} \mu_a^4$. This is independent of L and depends on M only through its eigenvalue spectrum. NM is maximized at $M = I_K$; the SM structure (3,2,1) with μ_w, μ_c < 1 gives NM < √3: P-indivisibility does not select the gauge group. Verified numerically for K = 1–12, L = 8–50, μ ∈ [0,1] with machine-precision agreement.